

Object-Oriented PHP Answers

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Part A // Class and Object

```
<?php class Student {  
    function sayHello(){  
echo "Hello Iam a student";  
}  
}  
  
$student1 = new Student(); $student->sayHello();  
  
>
```

Output Hello! I am a student

Part B Class with Constructor

```
<?php class Student {  
    public $name;    public $studentId;    public $department;  
    function __construct($name, $studentId, $department)    {  
$this->name = $name;  
        $this->studentId = $studentId;        $this->department = $department;    }  
    function showInfo()    {  
echo "Name: " . $this->name . "<br>"; echo "Student ID: " . $this->studentId . "<br>"; echo "Department: " .  
$this->department . "<br>"; }  
}  
  
$student1 = new Student("Ahmad", 1001, "Computer Science"); $student1->showInfo();  
  
>
```

Part C Create Another Object

```
$student2 = new Student("Sara", 1002, "Information Systems");  
  
$student2->showInfo();
```

Answer: Number of classes created = 1 (Student).

Answer: Number of objects created = 2 (\$student1 and \$student2).

Part D Access Modifiers

```
<?php class BankAccount {  
public $ownerName; private $balance;  
    function __construct($ownerName, $balance)    {  
$this->ownerName = $ownerName; $this->balance = $balance; }  
    function showBalance()    {  
echo "Balance: " . $this->balance; }  
}  
  
$account1 = new BankAccount("Ahmad", 5000); echo "Owner: " . $account1->ownerName . "<br>"; $account1->show  
>  
  
Question: Does echo $account1->balance; work?
```

No.

Why?

Because \$balance is declared private. A private property can only be accessed from inside the same class. It must be accessed through a public method such as showBalance().

Part E Inheritance

```
<?php class Person {  
    public $name;  
    function __construct($name)    {        $this->name = $name;    }  
    function introduce()    {  
        echo "My name is " . $this->name . "<br>"; }  
    class Student extends Person {  
        function study()    {  
            echo $this->name . " is studying."; }  
        }  
        $student1 = new Student("Ahmad");  
        $student1->introduce(); $student1->study();  
    }  
}
```

Part F Inheritance

Question 1: Person is the parent class

Question 2: Student is the child class

Question 3: The keyword is extends

Question 4: The introduce() method was written inside the Person class

Question 5: Yes

Question 6: Because Student extends Person, it inherits the public properties and methods of Person. Therefore, a Student object can call introduce().

Part G Vehicle and Car

```
<?php class Vehicle {  
    protected $brand;  
    function __construct($brand)    {        $this->brand = $brand;    }  
    function start()    {  
        echo "The vehicle is starting.<br>"; }  
    class Car extends Vehicle {  
        function showBrand()    {  
            echo "Car brand: " . $this->brand; }  
        }  
        $car1 = new Car("corrollah");  
        $car1->start(); $car1->showBrand();  
    }  
}
```

Output: The vehicle is starting. Car brand: corrollah

Complete Submission File

```
<?php  
// Part A, B, C: Student class Student {    public $name;    public $studentId;    public $department;
```

```

        function __construct($name, $studentId = null, $department = null)    {
$this->name = $name;
        $this->studentId = $studentId;          $this->department = $department;    }
        function sayHello()    {
echo "Hello! I am a student.<br>"; }
        function showInfo()
{
echo "Name: " . $this->name . "<br>"; echo "Student ID: " . $this->studentId . "<br>"; echo "Department: " .
$this->department . "<br><br>"; }
}
        $student1 = new Student("Ahmad", 1001, "Computer Science");
        $student1->sayHello();
        $student1->showInfo();
        $student2 = new Student("Sara", 1002, "Information Systems");
        $student2->showInfo();

// Part D: BankAccount class
class BankAccount {
public $ownerName; private $balance;
        function __construct($ownerName, $balance)    {
$this->ownerName = $ownerName; $this->balance = $balance; }
        function showBalance()    {
echo "Balance: " . $this->balance . "<br><br>"; }
}
        $account1 = new BankAccount("Ahmad", 5000); echo "Owner: " . $account1->ownerName . "<br>"; $account1->showBalance();

// Part E and F: Inheritance class
class Person { public $name;
        function __construct($name)    {          $this->name = $name;    }
        function introduce()    {
echo "My name is " . $this->name . "<br>"; }
        class StudentPerson extends Person {          function study()    {
echo $this->name . " is studying.<br><br>"; }
}
        $student3 = new StudentPerson("Ahmad");
        $student3->introduce();
        $student3->study();

// Part G: Vehicle and Car class
class Vehicle { protected $brand;
        function __construct($brand)    {          $this->brand = $brand;    }
        function start()    {
echo "The vehicle is starting.<br>"; } }
        class Car extends Vehicle {          function showBrand()    {
echo "Car brand: " . $this->brand; }
}
        $car1 = new Car("Toyota");
        $car1->start();

```

\$car1->showBrand();

?>